A Survival-Analysis Framework for Predicting and Explaining Time-to-Employment for Final-Year IT Undergraduates in Sri Lanka

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The problem: no graduate can see how long their own search will take

17,000

IT graduates a year in Sri Lanka

1 in 4

still searching at the survey: 'censored', not 'failed'

None found

in the literature review: a tool estimating WHEN a Sri Lankan IT graduate finds a first IT job

- **Gap**: existing tools classify or match jobs, and discard graduates who are still searching
- **Why survival analysis**: it estimates when, and uses people who have not yet found a job
- **What was missing**: individual explanations, linked to what employers actually advertise

Objectives and research questions

O1 / RQ1	Which factors are associated with time to employment?
O2 / RQ2	How well do profiles align with employer requirements and job-market demand?
O3 / RQ3	How accurately can a survival-analysis framework predict time to employment, and how do the models compare?
O4 / RQ4	Does the prototype give accurate, explainable, actionable guidance (usability and a benchmark against a general LLM)?

Method: three data sources, a pre-set evaluation design

1,175

graduate responses (885 modelled)

245

employer responses (239 analysed)

324

real entry-level IT job ads (211 employers)

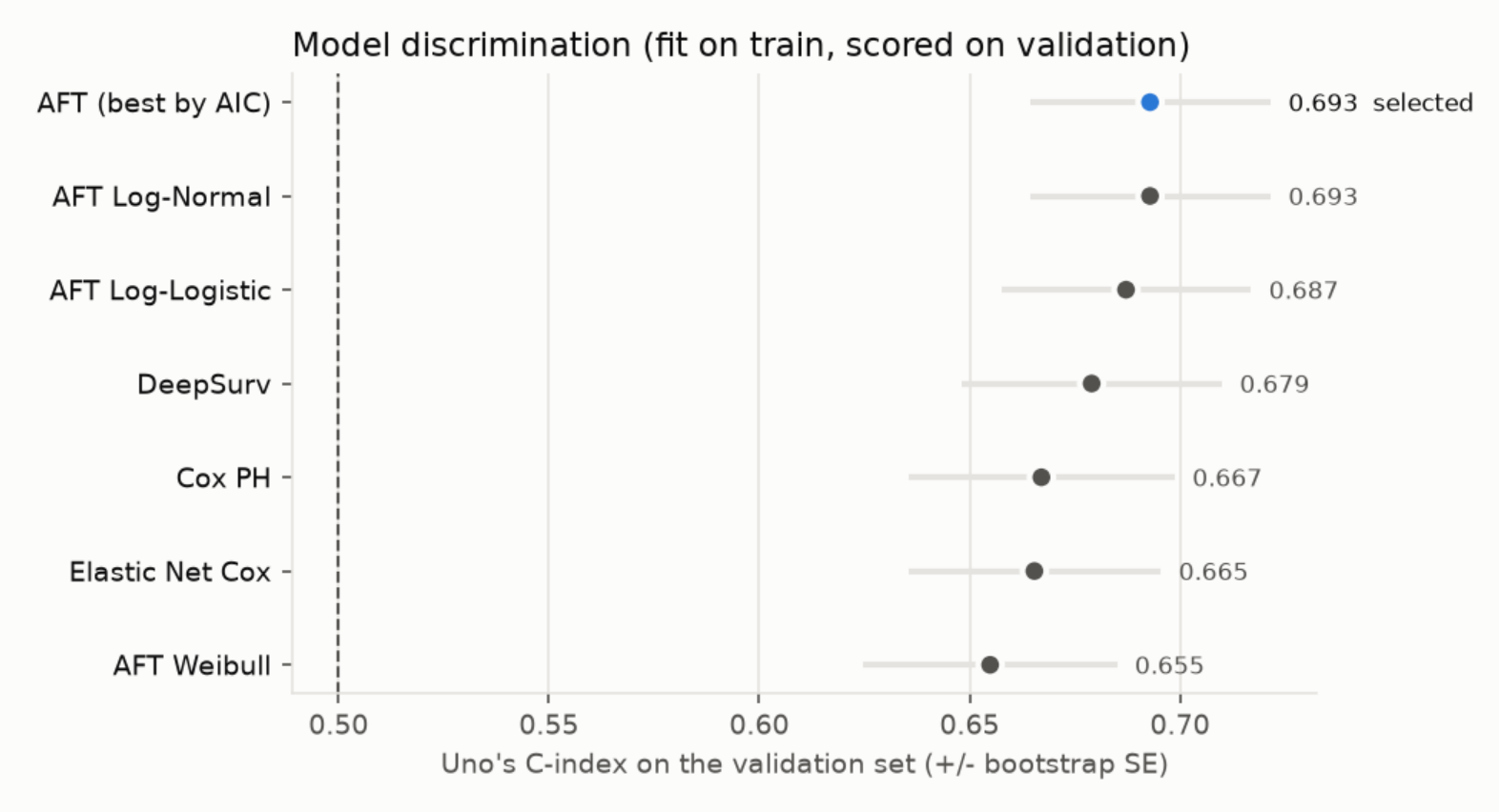
- **Approach**: pragmatist, deductive, quantitative; non-probability sampling (stated limitation)
- **Sample size**: Cochran n = 384 (95%, ±5%) → target about 400; collected well above it
- **Questionnaire quality**: 6 expert reviewers, pilot, KMO, parallel analysis; Cronbach's alpha and omega
- **Ethics**: consent on every response, no names or contacts, PDPA 2022, ethical approval before collection

Evaluation design: the test set was used once

620	133	132
training: fit and tune (cross-validation)	validation: choose the model by a pre-set rule	test: used once, never re-tuned

- **Models**: Cox PH, Elastic Net Cox, AFT (Weibull / log-normal / log-logistic), DeepSurv; no forests or ensembles
- **Primary metric**: Uno's C-index (built for censored data), with the integrated Brier score and calibration
- **Safeguards**: stratified split, preprocessing learned on training only, 29.2 events per parameter

Which model? AFT Log-Normal, chosen by the pre-set rule



- **Validation C**: 0.693 (0.636-0.745)
- **Brier score**: 0.171
- **Rule fixed**: before looking at results
- **More complex models**: did not do better

Test result: moderate ranking, honest reporting

0.619

Uno's C on unseen graduates (95% CI 0.557-0.683)

~62 / 100

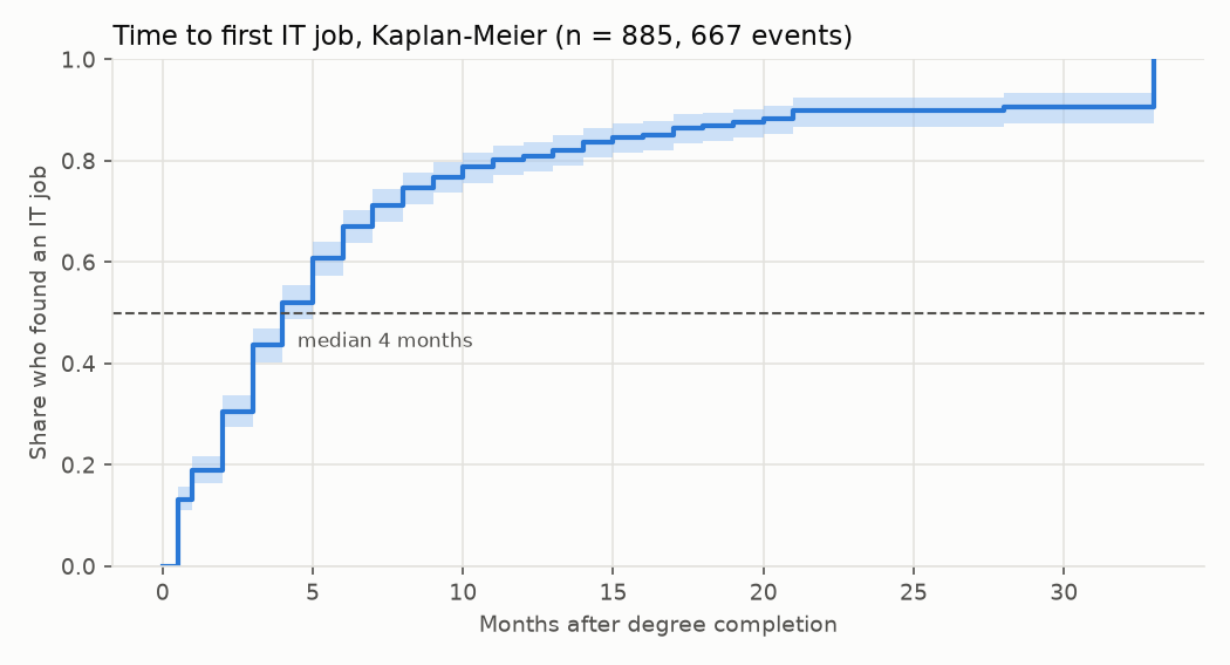
pairs ranked correctly (Harrell's C 0.626)

up to 17 pts

under-prediction of employment: probabilities are pessimistic

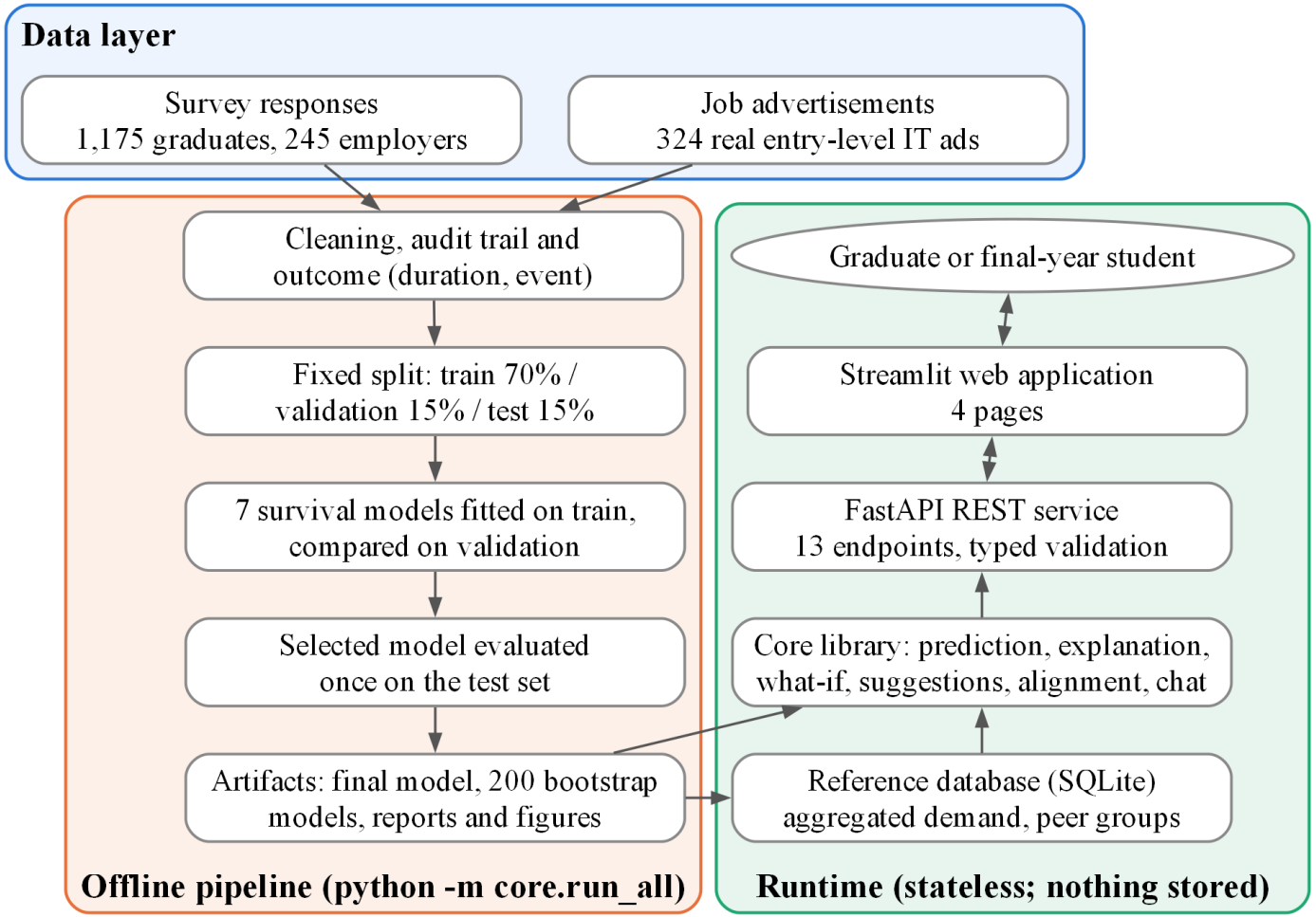
- **Reading it**: the ranking works; the exact probabilities are too pessimistic
- **Not better than Kaplan-Meier on error**: IBS 0.167 vs 0.164
- **Why**: small sets, faster hiring by later cohorts, one reference survey date
- **What the app does**: shows a range first and labels probabilities as indicative

What matters for finding work sooner



- **Half found a job by 4 months**; 67% by 6, 81% by 12
- **GPA and internships**: about 20% and 16% shorter search per SD
- **No clear difference**: by institution type or degree programme
- **Skill gap**: self-ratings below employer importance on 25 of 26 skills (cloud, testing, DevOps)
- Associations, not causes

The artifact: from profile to range, explanation and skill gaps



- **Output**: most likely time, typical range, confidence
- **Explains**: what helps and what holds you back
- **What-ifs**: associational wording only
- **Skill gaps**: against 324 real job ads
- **Nothing stored**: no profile kept; 295 tests pass

Why not just use ChatGPT? I tested it

0.619

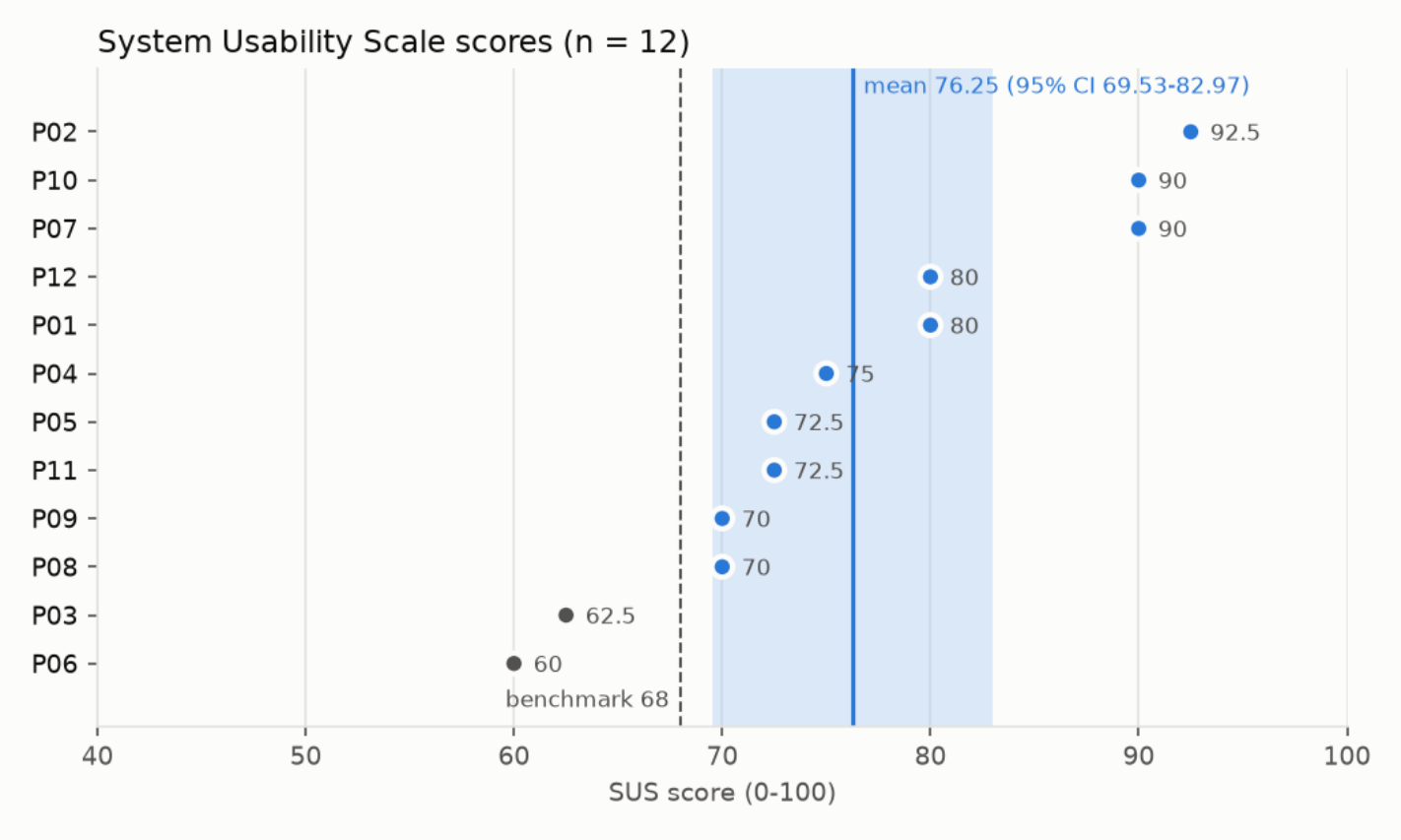
this model: Uno's C on the same 132 test graduates

0.539

ChatGPT: close to chance (difference 0.081, 95% CI 0.030-0.131)

- **ChatGPT**: over-optimistic, and gave different answers to the same profile on repeated runs
- **This system**: trained on 620 real local graduates, handles people still searching, consistent
- **Fair view**: ChatGPT is better at general advice and wording

Usability: SUS 76.3 with 12 users



- **SUS 76.3**: (95% CI 69.5-83.0) vs benchmark 68
- **10 of 12**: scored 68 or more
- **Feedback**: less on one screen, explain terms
- **Acted on**: reading guide, source line, collapsible chat
- **Honest**: peers, n = 12, fixes not re-tested

Limitations, stated openly

- **Sample**: non-probability; about a third from one institution; voluntary responders
- **Data**: self-reported outcome with recalled dates; one reference survey date
- **Model**: small evaluation sets (about 130); probabilities too pessimistic
- **Inference**: observational, so associations, not causes
- **Employer ratings**: cluster at 4-5, so used to rank skills, not as absolute levels
- **Usability**: 12 peer participants; task completion not recorded task by task

Contribution and conclusion

The first explainable survival-analysis system that predicts WHEN Sri Lankan IT graduates find a first IT job, triangulated with employer and job-ad data, and tested against a general LLM.

- **RQ1**: GPA and internships are associated with shorter searches
- **RQ2**: graduates rate themselves below employer expectations on 25 of 26 skills
- **RQ3**: moderate ranking (C 0.619); AFT Log-Normal chosen by a pre-set rule
- **RQ4**: usable (SUS 76.3) and better than ChatGPT (0.619 vs 0.539)
- **Next**: more institutions, real survey dates, recalibration, re-test the updated app

Thank you - questions?

Code and walkthrough notebook: github.com/ShaznaSalman/final-research-employability